

Question 4:

Deep dive into one design. This means describe in more detail how the new design would be built and tested to verify the new performance target. Clearly show your work!

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Sienna's Response Q3

The objective of this redesign is to improve the energy efficiency of a laptop during everyday web browsing by reducing processor operating temperature without increasing system complexity or requiring additional active cooling. Rather than adding larger fans or additional cooling hardware, this design reimagines the laptop chassis as a multifunctional structural and thermal component. I believe efficiency comes from using every component of a device to its fullest. Rather than treating the aluminum chassis as only a structural enclosure, I questioned whether it could also become a thermal management system that routes heat toward areas users rarely touch while maintaining the thin form factor expected of modern laptops.

Current cooling in the Microsoft Surface Laptop Studio2 is highly localized. Heat generated by the CPU and GPU is transferred through a thermal interface material into copper heat pipes, which transport the heat to compact fin stacks where dual blower fans force air through the fins and reject the heat into the surrounding environment. Although premium laptops such as Apple's fanless MacBook Air already use graphite sheets and the aluminum chassis to assist with passive heat spreading, the enclosure primarily functions as a thermal reservoir. User comfort is prioritized over maximizing passive heat rejection, leaving much of the chassis thermally underutilized.

Instead of redesigning the cooling system itself, I chose to redesign the chassis. My proposed design transforms the existing aluminum enclosure into a selective thermal architecture. The existing active cooling mechanisms remain unchanged and continue transporting heat away from the CPU. Rather than terminating at the conventional heat sink or fan, however, heat now also dissipates into a laser-patterned pyrolytic graphite sheet.

Pyrolytic graphite was selected because it already exists in modern consumer electronics and possesses an exceptionally high in-plane thermal conductivity, typically between 1000 and 1700 W/m·K, while remaining extremely thin and lightweight. Currently, Apple uses graphite as one large rectangular sheet that spreads heat relatively uniformly throughout the chassis. Rather than inventing a new material, I propose laser cutting the graphite into a branching thermal network that functions like a printed circuit board, routing heat instead of electricity.

Wide graphite branches with very low thermal resistance connect the processor to designated radiator regions located along the rear structural beam immediately in front of the display hinge, and the left and right side rails. These are regions that are naturally exposed to ambient air but rarely experience prolonged contact with the user's skin. High-contact regions, including the palm rests and bottom chassis, intentionally receive no direct graphite pathways. Instead, any heat reaching these areas must travel only through the aluminum chassis itself, which has a significantly lower thermal conductivity than the graphite network. Rather than actively moving heat toward specific regions, the design simply creates a path of much lower thermal resistance, allowing the majority of heat to naturally flow toward desired locations.

In order to prevent heat directed to radiator regions from traveling through the chassis to high user-contact areas, I first thought of dividing the chassis into multiple pieces separated by insulating materials. While this would reduce heat transfer, it conflicts with the seamless unibody construction used by premium laptops and would increase manufacturing complexity while reducing perceived product quality.

Instead, I looked toward thermal engineering. Since heat naturally follows the path of least thermal resistance, I suggest manipulating the geometry of the chassis itself. Rather than separating the laptop into multiple components, I propose machining internal thermal necks between the designated 'hot' and 'cool' regions. These thermal necks are created by locally reducing the internal cross-sectional area of the aluminum while leaving the exterior appearance completely unchanged. According to Fourier's Law of heat conduction, reducing the conduction area increases thermal resistance, slowing the movement of heat between regions without introducing additional materials or visible seams. From the outside, the laptop still appears as one continuous aluminum unibody, but internally the chassis functions as a thermally engineered network.

To reject the heat once it reached the designated radiator regions, my first instinct was to simply add ventilation holes. However, I found that manufacturers generally avoid unnecessary openings because they create pathways for dust, moisture, and debris to enter the computer. Rather than exposing the internals, I shifted toward increasing the effective surface area of the radiator regions themselves.

The underside of the rear structural beam and side rails would incorporate shallow CNC-machined microchannels approximately 0.3–0.5 mm deep. Unlike traditional heat sink fins, these microchannels remain hidden beneath the laptop, preserving the clean design while significantly increasing surface area available for natural convection. The rear structural beam would also be raised approximately 0.5–1.0 mm above the surrounding chassis, creating a passive chimney that enhances natural convection behind the display without additional fans or ventilation openings.

To further improve passive heat rejection, radiator regions would receive a high-emissivity hard-anodized coating. Polished aluminum has an emissivity of approximately 0.05, radiating heat relatively poorly. Hard-anodized aluminum, however, exhibits emissivities approaching 0.85–0.90, allowing the same component to reject considerably more heat through thermal radiation while maintaining corrosion resistance and durability.

This design contains no completely new cooling technology. Instead, existing technologies are reorganized into a different thermal architecture. The primary heat transport layer remains. The laser-patterned graphite sheet becomes the thermal routing layer, intentionally directing some heat toward preferred regions. Finally, the aluminum chassis becomes the distributed heat rejection layer, rejecting heat through strategically engineered radiator zones rather than treating the entire enclosure equally.

The concept could first be validated without building a completely new device by modifying an existing laptop. A replacement bottom chassis could be CNC machined with the proposed thermal necks, rear microchannels, and structural beam geometry while leaving the motherboard, processor, and existing cooling hardware unchanged. Commercial pyrolytic graphite sheets could then be laser cut according to the proposed thermal routing network and bonded to the existing copper heat pipe assembly using commercially available thermally conductive adhesive. This isolates the chassis and parallel heat routing as the only modified variables.

ANSYS Icepak simulations would compare heat flow, processor temperature, and chassis temperatures between the stock and redesigned devices under identical web browsing workloads. Following simulation, the physical prototype would be evaluated experimentally using identical workloads while logging CPU package power, processor temperature, clock frequency, and fan speed through HWiNFO64. A thermal camera can visualize how heat spreads across the chassis, verifying whether the graphite network successfully redirects heat toward the rear structural beam and side rails while maintaining comfortable temperatures beneath the user's hands.

The primary performance target is to reduce average processor operating temperature by approximately 5–10°C during sustained web browsing while maintaining palm rest temperatures below approximately 35°C and bottom chassis temperatures below approximately 40°C. The rear structural beam and side rails may safely operate at higher temperatures because they experience minimal user interaction. If successful, the processor should experience less thermal throttling, reduced fan usage, and complete computational tasks more quickly before returning to low-power idle states. Rather than simply making the laptop cooler, this design fundamentally changes how heat moves throughout the device. The chassis transitions from a passive structural enclosure into a multifunctional thermal management system that simultaneously improves cooling performance, energy efficiency, and user experience.

Thermal Zoning and Target Temperatures

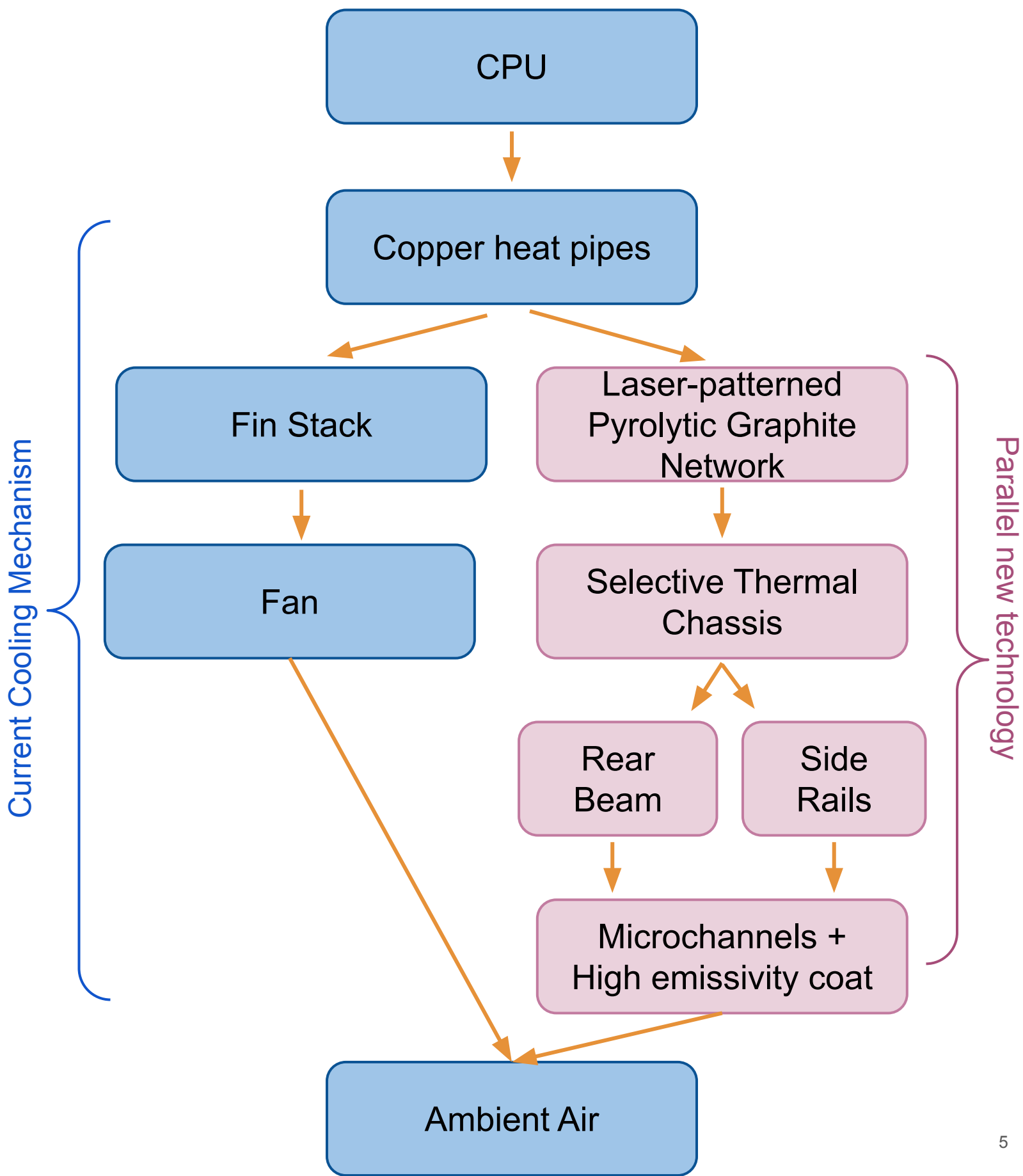


-  **CPU Die**
< 85 °C
-  **Rear Hinge (Max)**
< 50 °C
-  **Side Edges (Max)**
< 45 °C
-  **Palm Rests (Max)**
< 35 °C
-  **Keyboard (Max)**
< 38 °C
-  **Bottom Center (Max)**
< 40 °C

All temperatures based on 22 °C ambient, 30 W CPU load
Target complies with ISO 13732-1 skin contact guidelines

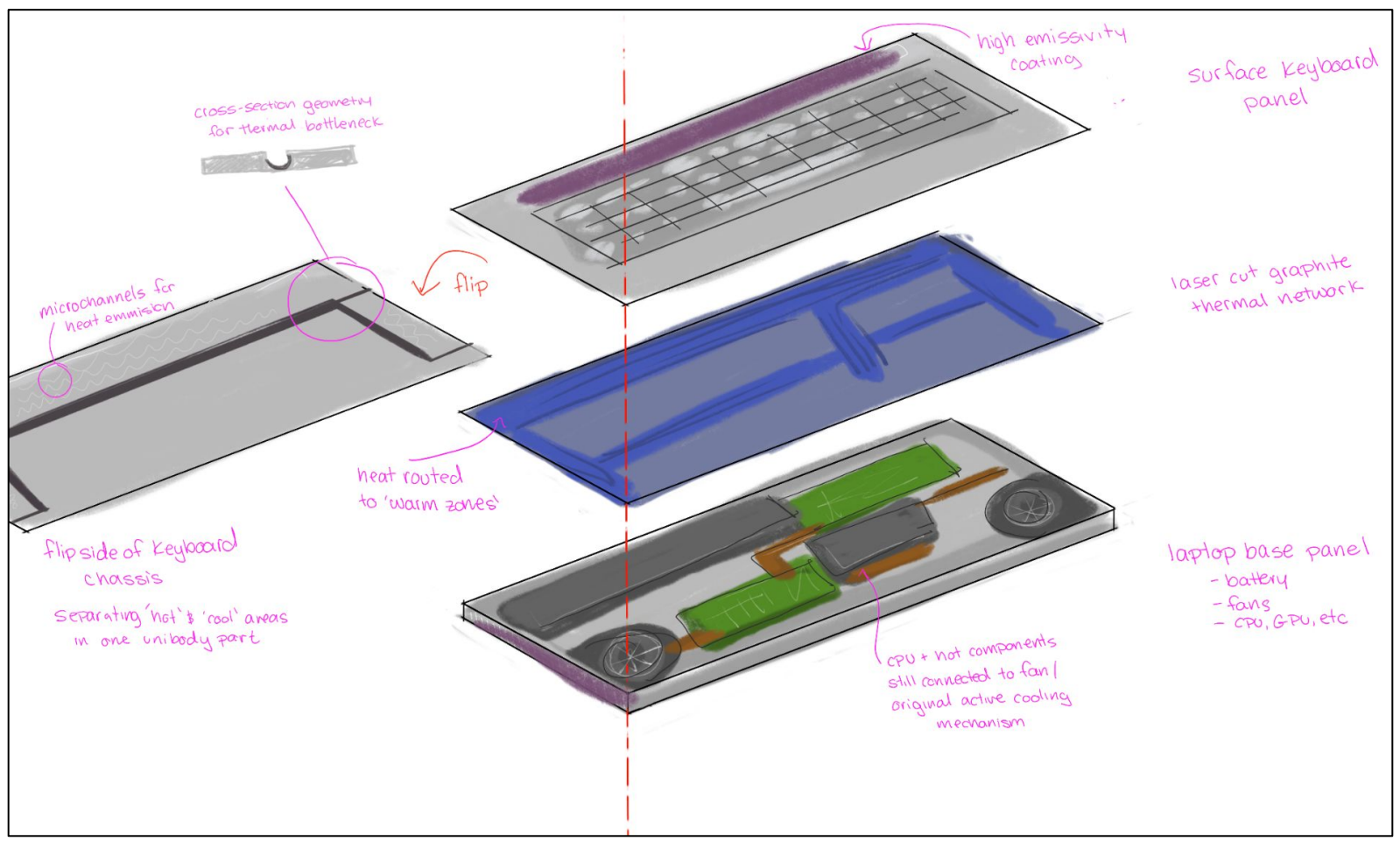
Thermal Architecture for Surface Studio 2:

Flow chart shows transfer of heat



Sketch of proposed design

Flow chart shows transfer of heat



Heat traveling through layers:

